

2 Weighted Voting

2.1 Beginnings

Definition 2.1.1 Each individual which is casting a vote is called a **player**, which we will denote $P_1, P_2, \dots, P_N$, where N is the number of players.

Each player is given a **weight**, which usually represents how many votes they get.

The **quota** is the minimum weight needed for the proposal to be approved.

We will often represent all of this information in the following form:

$$[q : w_1, w_2, \dots, w_N]$$

where q represents the quota, and w_i is the weight of player P_i .

Example 2.1.2 Suppose that in a small company, There are four shareholders, the first has a 30% ownership, the second and third both have a 25% ownership, and the fourth a 20% ownership. The company by-laws state that more than 50% of the ownership must approve any decision to open a new location. Represent this situation as a wieghted voting system.

$[51 : 30, 25, 25, 20]$

Example 2.1.3 What is wrong with the following weighted voting system?

$$[11 : 4, 3, 2, 1]$$

This is gridlock, nothing can be approved.

Example 2.1.4 What is wrong with the following weighted voting system?

$$[3 : 3, 2, 1]$$

This is anarchy, since both yes and no could reach the quota simultaneously.

2.1.1 Power

Example 2.1.5 Consider the weighted voting system

$$[8 : 9, 3, 2, 1].$$

Where does the power lie? Why?

Player 1 has all the power.

Definition 2.1.6 A player is said to have **veto power** if they are necessary for a proposal to pass.

Definition 2.1.7 A **dictator** is a player who can reach the quota by themselves and also has veto power.

Example 2.1.8 Determine all the players who have veto power in the weighted voting system

$$[22 : 8, 6, 4, 4, 3, 2, 1].$$

In the previous example, player 1 is a dictator.

How do we determine who has veto power? Derive the formula

$$V - q < w_i.$$

Example 2.1.9 Determine all the players who have veto power in the following weighted voting system:

$$[34 : 12, 11, 10, 7, 3].$$

Players 1, 2, and 3 all have veto power.

Definition 2.1.10 A player is called a **dummy** if their vote is never essential for a group to reach quota.

Example 2.1.11 Identify any dummies in the weighted voting system

$$[10 : 7, 6, 2].$$

Before we can define power, we need a few more definitions.

Definition 2.1.12 A **coalition** is a group of players voting the same way.

Definition 2.1.13 A coalition is said to be a **winning coalition** if the players in the coalition reach the quota.

Definition 2.1.14 A player in a coalition is said to be critical if them leaving the coalition would change it from a winning coalition to a losing coalition.

Example 2.1.15 Consider the weighted voting system:

$$[14 : 7, 6, 3, 3].$$

Identify all winning coalitions.